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Review

# Plant Immunity: Danger Perception and Signaling

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Plants employ numerous cell-surface and intracellular immune receptors to perceive a variety of immunogenic signals associated with pathogen infection and subsequently activate defenses. Immune signaling is potentiated by the major defense hormone salicylic acid (SA), which reprograms the transcriptome for defense. Here we highlight recent advances in understanding the mechanisms underlying activation of the main classes of immune receptors, summarize the current understanding of their signaling mechanisms, and discuss an updated model for SA perception and signaling. In addition, we discuss how different receptors are organized into networks and the implications of such networks in the integration of complex danger signals for appropriate defense outputs.

## Introduction

Terrestrial plants are naturally associated with numerous microbial organisms including viruses, bacteria, fungi, and oomycetes (Teixeira et al., 2019). Unlike commensal or beneficial microbes, pathogenic microbes can not only deprive nutrients but also perturb physiological processes, arrest growth, and inflict tissue damage to their hosts via production of toxins, cell-wall-degrading enzymes, and virulence proteins (Dou and Zhou, 2012). A major challenge of land plants is to distinguish pathogens from commensals or beneficial microbes and to appropriately activate defenses when under attack by pathogens.

Higher plants possess an enormous number of cell-surface and intracellular immune receptors to perceive a variety of immunogenic signals associated with pathogen infections (Kourelis and van der Hoorn, 2018; Van de Weyer et al., 2019). Cell-surface immune receptors consists of receptor-like proteins (RLPs) and receptor kinases (RKs), which are conceptually analogous to Toll-like receptors in animals (Kawai and Akira, 2010; Tang et al., 2017). Plant intracellular immune receptors are nucleotide-binding, leucine-rich repeat receptors (NLRs), which also exist in animals (Jones et al., 2016). In plants, cell-surface immune receptors are responsible for the perception of microbe- or host-derived immunogenic molecular patterns and are referred to as pattern recognition receptors (PRRs), whereas NLRs specifically sense the more variable pathogen effector proteins that are delivered into the plant cell.

In the last decade, major progress has been made in deciphering plant immune signaling governed by both PRRs and NLRs, particularly in *Arabidopsis*. This Review summarizes our current understanding concerning how activation of immune receptors leads to major downstream signaling events, such as calcium influx, a burst of reactive oxygen species (ROS), and production of immunogenic peptides and defense hormones. We highlight recent breakthroughs that have substantially advanced our understanding of how NLRs are activated to initiate defense signaling. In addition, we discuss how different immune receptors form surveillance networks to integrate different immunogenic signals that could be the key to interpret threat of pathogen attacks for effective defenses. Furthermore, we highlight recent breakthroughs in the biosynthesis and perception of the major defense hormone salicylic acid (SA) because of its critical roles in the potentiation of PRR- and NLR-mediated signaling.

## Immunogenic Signals and Microbial Recognition

Immunogenic signals that activate plant defenses come in diverse forms. Immunogenic microbial patterns that are relatively conserved within taxonomic groups are referred to as microbe-associated molecular patterns (MAMPs) (Ausubel, 2005). Plants use lytic enzymes to facilitate the release of MAMP epitopes from invading pathogens, as in the case of the bacterial flagellin epitope flg22 and peptidylglycan (Buscaill et al., 2019; Liu et al., 2014). In addition, pathogens can inflict damage to plant tissues via cell-wall-degrading enzymes and toxins, which releases plant cell wall fragments, extracellular ATP, and nicotinamide adenine dinucleotide (NAD) (Gust et al., 2017 and references therein). These plant-derived molecules can serve as damage-associated molecular patterns (DAMPs) to trigger immune responses. Furthermore, altered cell wall integrity is also actively monitored by plants and likely contributes to pathogen sensing (Bacete et al., 2018).

Plants also actively produce immunogenic peptides called phytocytokines that are perceived by PRRs or PRR-associated receptors to modulate immune responses in a way analogous to cytokines in animal immunity (Gust et al., 2017). Although our knowledge concerning phytocytokines is limited,

protease-mediated maturation has emerged as a major mechanism regulating the peptide accumulation in the apoplast. For instance, plant elicitor peptides (Peps) are prominent phytocytokines that trigger defenses through the leucine-rich repeat RKs (LRR-RKs) PEPR1 and PEPR2 (Yamaguchi et al., 2010). Multiple cytoplasmic type-II metacaspapses (MCs), which are calcium-dependent cysteine proteases, control the cleavage and maturation and release of Peps into the apoplast (Hander et al., 2019; Shen et al., 2019). Plants additionally use various apoplastic proteases to control the maturation of peptides such as the SA-inducing peptide Zip1 in maize (Ziemann et al., 2018) and *Arabidopsis* RAPID ALKALINIZATION FACTORs (RALFs), which regulate numerous biological processes including immune responses (Stegmann et al., 2017).

Pathogenic microbes secrete effector proteins into the plant cell or extracellular spaces to promote parasitism (Dou and Zhou, 2012). However, host-pathogen co-evolution can render these pathogen effectors immunogenic when the host plant acquires cognate immune receptors (Cui et al., 2015; Kourelis and van der Hoorn, 2018). MAMPs, DAMPs, and immunogenic apoplastic effectors are perceived by PRRs, whereas intracellular effectors are perceived by NLRs. Thus, any given infection involves perception of multiple danger signals by different immune receptors.

MAMPs exist not only in pathogens but also in commensal and beneficial microbes. As discussed in a recent review (Teixeira et al., 2019), commensal and beneficial microbes can actively suppress immune responses or evade detection by plant PRRs. However, normal MAMP perception also actively contributes to microbiome homeostasis, because mutations that simultaneously block MAMP perception and cause an abnormal aqueous microenvironment in the apoplast result in dysbiosis in the leaf (Chen et al., 2020). In animals, perception of MAMPs by innate immune receptors can lead to immune tolerance and establishment of long-term colonization of the commensals (Chu and Mazmanian, 2013). Although it remains unknown how MAMP perception promotes microbiome homeostasis in plants, a recent study suggests that establishment of mycorrhizal symbiosis in *Medicao truncatula* requires not only recognition of the mycorrhizal lipochitooligosaccharide, a well-known symbiotic signal molecule, but also perception of chitin oligomers (Feng et al., 2019). Whereas chitin oligomers alone can stimulate immune responses in *M. truncatula*, a combination of chitin oligomers and lipochitooligosaccharide synergistically enhances symbiosis. Although the underlying mechanism is unclear, this finding highlights the importance of the integration of multiple microbial signals for proper responses in the host plant.

A very recent and elegant study showed that differentiated *Arabidopsis* roots can integrate cell damage and MAMP signals to specifically activate defenses in response to pathogenic bacteria but not to commensal bacteria (Zhou et al., 2020). Differentiated roots express very low levels of PRRs and respond poorly to MAMPs, which allows optimum colonization by commensal bacteria. Infection by pathogenic bacteria, however, causes cell damage in these roots and induces the expression of multiple PRRs in the neighboring cells, allowing the latter to activate strong defenses upon the perception of MAMPs. Here, the combination of a wound signal and a MAMP is perceived by plants as a true danger signal.

Immune signaling from each receptor is often investigated as an independent process, and different immune receptors are considered functionally redundant. However, it is not clear whether a single immunogenic signal is sufficient to inform the host plant the nature of the threat. As discussed above, multiple signals including MAMPs, DAMPs, wounding, phytocytokines, and/or symbiotic signal molecules are perceived during the interaction with a single microbe, and integration of different signals could be critical for plants to distinguish friends from foes.

## Organization of Immune Receptor Networks at the Cell Surface

It is well-established that ligand perception by PRRs requires co-receptors (Tang et al., 2017 and references therein). For instance, flagellin sensing 2 (FLS2) and EF-TU receptor (EFR), which respectively are receptors for flg22 and bacterial elongation factor Tu epitope elf18, require the somatic-embryogenesis receptor-like kinase (SERK) family of LRR-RKs such as BRI1-associated kinase 1 (BAK1) as co-receptors. Similarly, lysin motif (LysM)-containing receptor kinase 5 (LYK5), a chitin receptor, requires another LysM-RK chitin elicitor receptor kinase 1 (CERK1) as a co-receptor.

The interaction of FLS2 with BAK1 is induced by flg22 and subjected extensive regulation by several accessory LRR-RKs carrying short LRR domains. Among these, BAK1-interacting RLK2 (BIR2) and BIR3 negatively regulate the FLS2-BAK1 interaction in the resting state by a sequestration of BAK1 (Halter et al., 2014; Imkampe et al., 2017). A systems study on 200 *Arabidopsis* LRR-RKs uncovered a comprehensive cell-surface interaction network ($CSI^{LRR}$) in which diverse LRR-RKs are connected through short LRR-RKs as hubs, including BAK1, APEX, and RLK902 (Smakowska-Luzan et al., 2018). The network additionally contains short LRR-RKs FLS2-interating receptor (FIR) and nuclear shuttle protein-interacting kinase 1 (NIK1). Importantly, APEX and NIK1 negatively regulate the flg22-induced FLS2-BAK1 interaction (Smakowska-Luzan et al., 2018; Li et al., 2019), whereas FIR promotes it (Smakowska-Luzan et al., 2018). A future challenge will be to disentangle how different accessory LRR-RKs are coordinated during MAMP perception and immune activation.

Besides the aforementioned co-receptors and accessory RKs, PRRs can interact with RKs carrying distinct ectodomains. The malectin-domain RK impaired oomycete susceptibility 1 (IOS1) constitutively interacts with FLS2, EFR, and CERK1 to positively regulate anti-fungal and anti-bacterial immunity by modulating PRR-co-receptor complex formation (Yeh et al., 2016). In addition, the *Cantharanthus roseus* receptor-like kinase-1-like (CrRLK1L) family RK FERONIA (FER) is required for ligand-induced FLS2-BAK1 and EFR-BAK1 interactions, likely by acting as a scaffold protein (Stegmann et al., 2017). Interestingly, two other CrRLK1L family members, ANXUR1 and ANXUR2, interact with FLS2 to negatively regulate FLS2-mediated antibacterial immunity, likely through an induced sequestration of BAK1 (Mang et al., 2017). Thus, a single PRR could recruit different scaffold proteins to either positively or negatively regulate its function.

RALFs are perceived by FER, which requires Lorelei-like glycosylphosphatidylinositol-anchored proteins (LLGs) as

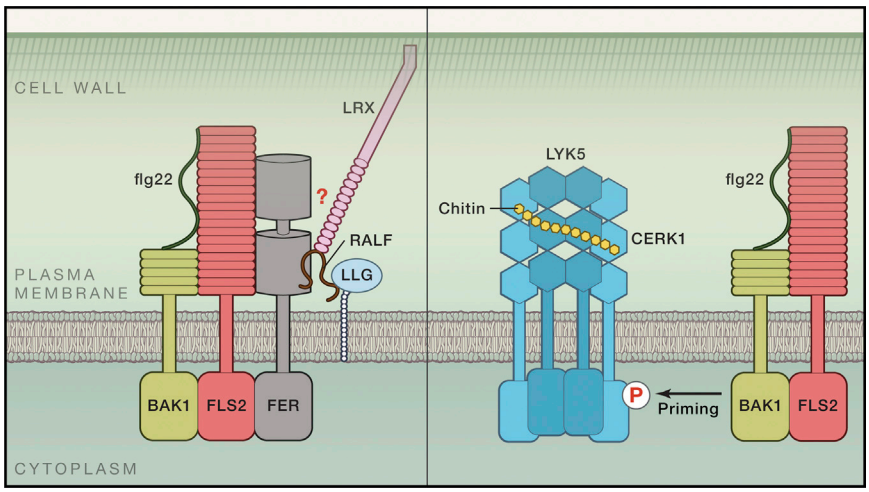



**Figure 1. Examples of PRR-RK Interactions for Signal Integration**

The interaction of the FLS2-BAK1 complex with the FER-LLG complex provides an example as to how a PRR can integrate multiple signals. The flg22-induced FLS2-BAK1 complex formation is promoted by FER, which functions as a scaffold protein. FER and its co-receptor LLGs can perceive a variety of RALF and LRX proteins, and the interaction of FER-LLG with the FLS2-BAK1 complex might enable integration of diverse signals, such as cell growth and the alteration of cell wall integrity, to further regulate immune signal output. The interaction between the FLS2-BAK1 complex and the LYK5-CERK1 complex provides an example where activation of one PRR primes a second PRR for increased signal output.

co-receptors (Xiao et al., 2019). In addition to acting as a scaffold protein, FER perceives RALF23 and RALF17 to negatively or positively regulate the immune outputs of FLS2 and EFR (Stegmann et al., 2017). Notably, FER is known to function in cell-wall-integrity sensing, a process mediated by RALFs and leucine-rich extensins (LRXs) (Dünser et al., 2019). Because RALFs are important signals for diverse biological processes, the coupling of PRRs with FER could allow plants to integrate cell wall damage and growth signals for different defense outputs (Figure 1).

A recent report showed that activation of FLS2 by flg22 induces a phosphorylation on the juxtamembrane domain of CERK1 in a BAK1-dependent manner, which stabilizes CERK1 and enhances chitin-triggered antifungal immunity (Gong et al., 2019) (Figure 1). The finding provides a mechanistic explanation for bacterial-induced priming in antifungal immunity. It is interesting to note that IOS1 is required for defense priming in response to β-aminobutyric acid, although it remains unknown whether IOS1 is a receptor for unknown ligands (Yeh et al., 2016). Furthermore, NIK1 plays a role in antiviral immunity, and activation of FLS2 by flg22 induces phosphorylation of NIK1 to promote antiviral defense, suggesting an interplay between different immune signaling pathways (Li et al., 2019). The aforementioned studies indicate that the interactions of PRRs with different RKs provide an important mechanism for immune priming.

RKs regulating immunity, reproduction, growth, and development often recruit the same co-receptors, raising a question concerning how specificity of signal outputs is achieved. For instance, brassinosteroid-insensitive 1 (BRI1), an LRR-RK perceiving brassinosteroid and regulating plant growth, also uses BAK1/SERKs as co-receptors (See Tang et al., 2017 and references therein). Several specific phosphosites in BAK1 are required for anti-bacterial immunity but not brassinosteroid signaling, suggesting that different "phosphocodes" are used for different receptor-co-receptor pairs (Perraki et al., 2018). In addition, BRI1 and FLS2 are localized to different plasma membrane (PM) nanodomains, providing another potential layer of specificity control through spatial distribution (Bücherl et al., 2017). Future studies on the organization of various RKs in nanodomains will shed light on how PRRs integrate different danger signals to inform cells for proper response.

## Signaling from PRRs to Downstream Components

Activation of PRRs triggers a number of key signaling modules including rapid phosphorylation of receptor-like cytoplasmic kinases (RLCKs), an influx of calcium across PM, a burst of ROS, and activation of calcium-dependent kinases (CPKs), mitogen-activated protein kinase (MAPK) cascades, and heterotrimeric G proteins (Tang et al., 2017).

A number of RLCKs play critical roles in linking PRRs to key downstream signaling modules. In *Arabidopsis* these include botrytis-induced kinase 1 (BIK1) and some PBS1-like (PBL) kinases of RLCK subfamily VII, as well as brassinosteroid-signaling kinase 1 (BSK1) of the RLCK subfamily II (Tang et al., 2017 and references therein). Upon activation by PRRs, these RLCKs phosphorylate and activate downstream components. For instance, BIK1 directly phosphorylates the N terminus of NADPH oxidase respiratory burst oxidase homolog D (RbohD) to stimulate ROS (Figure 2) (see Kadota et al., 2015 and references therein). The MAMP-triggered ROS burst additionally requires calcium-dependent kinase 5 (CPK5)-dependent phosphorylation and direct binding of $Ca^{2+}$ to EF-hands in RbohD N terminus (Figure 2). Furthermore, a cysteine-rich receptor kinase was shown to phosphorylate RbohD C terminus and promote NADPH oxidase activity (Kimura et al., 2020).

MAMP-triggered calcium influx is of profound importance to plant immunity (Tang et al., 2017). The activation of calcium channels is dependent on BIK1 and PBL1 (Tang et al., 2017 and references therein). In *Arabidopsis*, cyclic nucleotide-gated channel proteins CNGC2 and CNGC4 were recently shown to form a heteromeric calcium-permeable channel required for flg22-triggered calcium influx, provided that plants are grown under high external calcium concentration (Tian et al., 2019). The CNGC2/4 channel is kept in a resting state by the calmodulin CAM7. BIK1 was shown to phosphorylate CNGC4 at several sites required for the activation of the channel (Figure 2). Notably, CNGC2/4 are also required for $H_2O_2$-induced calcium flux and

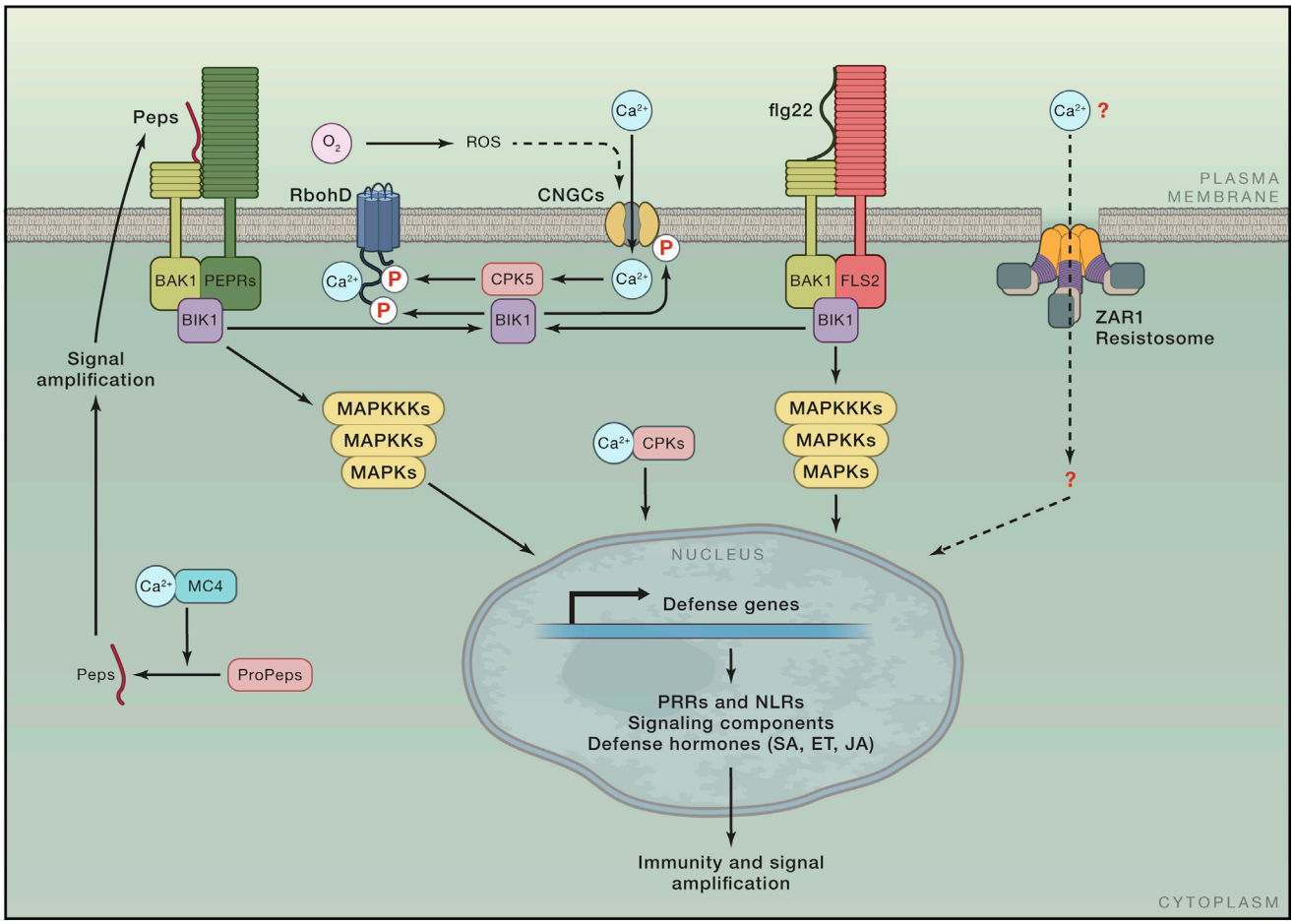



**Figure 2. Major Signaling Events Downstream of Representative PRRs and CNL Proteins**

FLS2 and PEPR are shown as representative PRRs perceiving MAMPs and phytocytokines, respectively. The ZAR1 resistosome on the PM is shown as a representative CNL with a hypothetical role as an ion channel. PRRs activate downstream components primarily via RLCKs, represented by BIK1. Under high external calcium concentrations, BIK1-mediated phosphorylation of CNGC2/4 contributes to MAMP-triggered calcium influx. BIK1 and CPK5 phosphorylate RbohD for the activation of a ROS burst, and phosphorylation of MAP kinase kinase kinases (MAPKKKs) by BIK1 and/or related RLCKs activate MAPK cascades. Not shown in the figure are the unknown calcium channels downstream of PRRs, heterotrimeric G proteins, and other regulatory components. Increased cytosolic calcium could lead to activation of calcium-dependent MCs, which can cleave protein precursor ProPeps to release mature Peps into the apoplast, providing a mechanism of signal amplification and crosstalk. Calcium signal can also be decoded by CPKs which together with MAPKs regulate transcription of defense genes encoding PRRs, NLRs, immune signaling proteins, and biosynthetic enzymes of defense hormones, providing another mechanism of signal potentiation between PRRs and NLRs. Solid arrows are supported by literatures, whereas dashed ones are hypothetical.

flg22-induced ROS burst (Tian et al., 2019), which is consistent with the notion that ROS and calcium signals mutually amplify each other (Kadota et al., 2015). An independent study showed that the rice OsCNGC9 is a calcium-permeable channel required for chitin-triggered cytosolic calcium influx and resistance to the rice blast fungus (Wang et al., 2019b). OsCNGC9 can be similarly phosphorylated by a rice RLCK VII isoform RLCK185, which increases the channel activity in a heterologous system. Although these studies are key advances in PRR signaling, outstanding questions remain. For example, BIK1 and PBL1 are required for calcium influx in plants grown under normal external calcium (Li et al., 2014), which differ from the function of CNGC2/4 (Tian et al., 2019). Furthermore, *cngc2* and *cngc4* mutants are only partially affected in flg22-induced calcium influx, suggesting the presence of additional calcium channels. Given that plants possess multiple types of calcium channel proteins (Tang et al., 2017), it remains a major task to fully understand the nature of calcium channels governing pattern-triggered calcium flux.

The activation of MAPK cascades is another major early signaling event required for the defenses activated by PRRs (Tang et al., 2017). One cascade employs the MAPKKK MEKK1 for the activation of downstream MAPKs MPK4 and MPK11, and the other activates MPK3 and MPK6 through redundant upstream kinases MAPKKK3 and MAPKKK5 (Bi et al., 2018; Sun et al., 2018; Yamada et al., 2016; Yan et al., 2018). Multiple RLCK VII members directly operate downstream of CERK1, and likely other PRRs, to phosphorylate the C termini of MAPKKK5 and MEKK1 and activate the two cascades (Bi et al., 2018). Interestingly, BSK1 can phosphorylate a site in the N terminus of MAPKKK5, and this site is required for disease resistance

function of MAPKKK5 (Yan et al., 2018). MAPKKK5 and MEKK1 are additionally phosphorylated in the C termini by MPK3/6 and MPK4, respectively, indicating a feedback regulation (Bi et al., 2018).

Unlike their animal counterparts, plant heterotrimeric G proteins are not coupled to seven-transmembrane G protein coupled receptors for function. In *Arabidopsis*, several heterotrimeric G protein subunits interact with FLS2 and BIK1 in the resting state (Liang et al., 2016, 2018). The regulator of G protein signaling 1 (RGS1), which accelerates GTPase activity of Gα to negatively regulate G protein activity, also associates with the FLS2-G protein complex to maintain the G proteins in a resting state (Liang et al., 2018). Upon flg22 stimulation, BIK1 phosphorylates RGS1 to destabilize its interactions with FLS2 and Gα proteins, allowing the G proteins to be activated. The non-canonical extra-large Gα protein XLG2 is required for the activation of RbohD and increased expression of defense genes (Liang et al., 2016, 2018). Future studies are needed to identify targets of these G proteins and the underlying regulatory mechanisms.

## General Features of NLRs and NLR-Mediated Defenses

Although both plant and animal NLRs share common structural features including a central nucleotide-binding domain (NBD) shared by APAF-1, R proteins, CED-4 (NB-ARC) or NAIP, CIITA, HET-E, TP1 (NACHT) domain, and a C-terminal LRR domain, they evolved independently (Jones et al., 2016). The plant lineage of NLRs first appeared in green algae and are classified into three major groups according to their N-terminal domains: TIR-NBS-LRR (TNL) proteins carrying a Toll Interleukin-1 Receptor (TIR) domain, CC-NBS-LRR (CNL) proteins carrying a coiled coil (CC) domain, and RPW8-NB-ARC-LRR (RNL) proteins, carrying a RPW8-like ($CC_R$) domain. Although TNLs and CNLs are fast evolving, RNLs are relatively conserved across different plant species and are divided into two subclasses, ADR1 and NRG1, each consisting of a few isoforms in any given plant species (Jubic et al., 2019). In general, the LRR domain functions in the recognition of effectors, although exceptions exist (Cui et al., 2015). The NBD binds ADP in the resting state and ATP in the active state. The TIR and CC domains function in signaling (Jubic et al., 2019). A distinct feature of NLR-triggered immunity is induction of a programmed cell death called hypersensitive response (HR), among other defense responses.

## Sensor NLRs Detect Effectors through Multiple Strategies

Most NLRs function in effector-recognition, and these are referred to as sensor NLRs. Some of them can recognize pathogen effectors directly via physical interaction. For example, the TNLs L5, L6, and L7 encoded by allelic genes of flax directly bind different variants of AvrL567 effectors from the rust fungus *Melampsora lini* to trigger disease resistance (Dodds et al., 2006).

More often, sensor NLRs sense the effector-induced steric alteration or post-translational modification on another host protein. Classical examples include the CNLs RPM1 and RPS2, which associate with the plasma-membrane-associated host protein RIN4 (RPM1-interacting 4) to monitor *Pseudomonas syringae* effectors AvrRpm1, AvrB, and AvrRpt2 (Axtell and Staskawicz, 2003; Mackey et al., 2002). Likewise, the CNL ZAR1 associates with RLCK XII family pseudokinases ZED1 and RKS1 to monitor the *P. syringae* effector HopZ1a and *Xanthomonas campestris campestris* effector AvrAC, respectively, and also trigger resistance (Lewis et al., 2013; Wang et al., 2015). AvrAC is an uridylyltransferase that modifies members of the RLCK VII including BIK1 and PBL2, and the modified PBL2 ($PBL2^{UMP}$) then interacts with the ZAR1-RKS1 complex for immune activation (Wang et al., 2015). The CNL SUMM2 indirectly senses the activity of *P. syrinage* effector HopAI1, a non-canonical MAPK phosphatase that inactivates MAP kinases (Zhang et al., 2017). SUMM2 directly interacts with CRCK3, a calmodulin-binding receptor-like kinase and a substrate of MPK4, to monitor homeostasis of MPK4 activity.

In some cases, an effector target coding sequence is fused to an NLR coding sequence to produce a single NLR protein with "integrated domain" (ID), and the modification of ID by effectors activates the NLR (Kourelis and van der Hoorn, 2018). For example, the TNL RRS1 carries a C-terminal WRKY domain, which is the DNA-binding domain of a major class of defense-related transcription factors (Le Roux et al., 2015; Sarris et al., 2015). The *Ralstonia solanacearum* effector PopP2, an acetyltransferase originally evolved to acetylate and inhibit WRKY transcription factors, also acetylates the WRKY domain in RRS1 to trigger immunity.

## Signaling from CNLs

The NB-ARC/NACHT domain in animal NLRs is a molecular switch that exists in an ADP-bound inactive state or ATP-bound active state. The binding of bacterial MAMPs converts the sensor NLR proteins NAIPs into an ATP-bound, activated form, which then interacts with another NLR protein NLRC4 to form oligomeric inflammasomes (Hu et al., 2013). The formation of inflammasomes aggregates the CARD domain that then recruits caspases to execute cell death and immune activation. A major advance in plant NLR research came with the recent cryoelectron microscopy (cryo-EM) structural studies of full-length ZAR1 protein in complex with RKS1 and $PBL2^{UMP}$ (Wang et al., 2019a; Wang et al., 2019c). The ZAR1-RKS1 complex is ADP bound and maintained in a resting state by intramolecular interactions that block ZAR1 oligomerization (Wang et al., 2019c). The binding of $PBL2^{UMP}$ to RKS1 indirectly disrupts the ADP binding of ZAR1. The ADP-depleted monomeric ZAR1 readily binds ATP to trigger drastic conformational changes in ZAR1, which unmasks ZAR1-ZAR1 interaction interfaces, forming a pentameric complex termed resistosome that largely resembles inflammasomes (Wang et al., 2019a). The CC domain in ZAR1 undergoes a major conformational switch so that the first α-helix transformed to form a barrel-like amphipathic structure that can insert into PM, and this PM-association is required for disease resistance and cell death conferred by ZAR1 (Figure 2). These structural data suggest that the ZAR1 resistosome directly forms a narrow pore in the PM to trigger immunity.

An outstanding question is whether the structural model is applicable to other NLRs in plants. A more recent study shows that a significant number of CNLs, including ZAR1, barley MLA, and Solanaceae-specific NLRs required for cell death (NRCs), share a conserved MADA motif in the first α-helix of the CC domain (Adachi et al., 2019). Several key hydrophobic

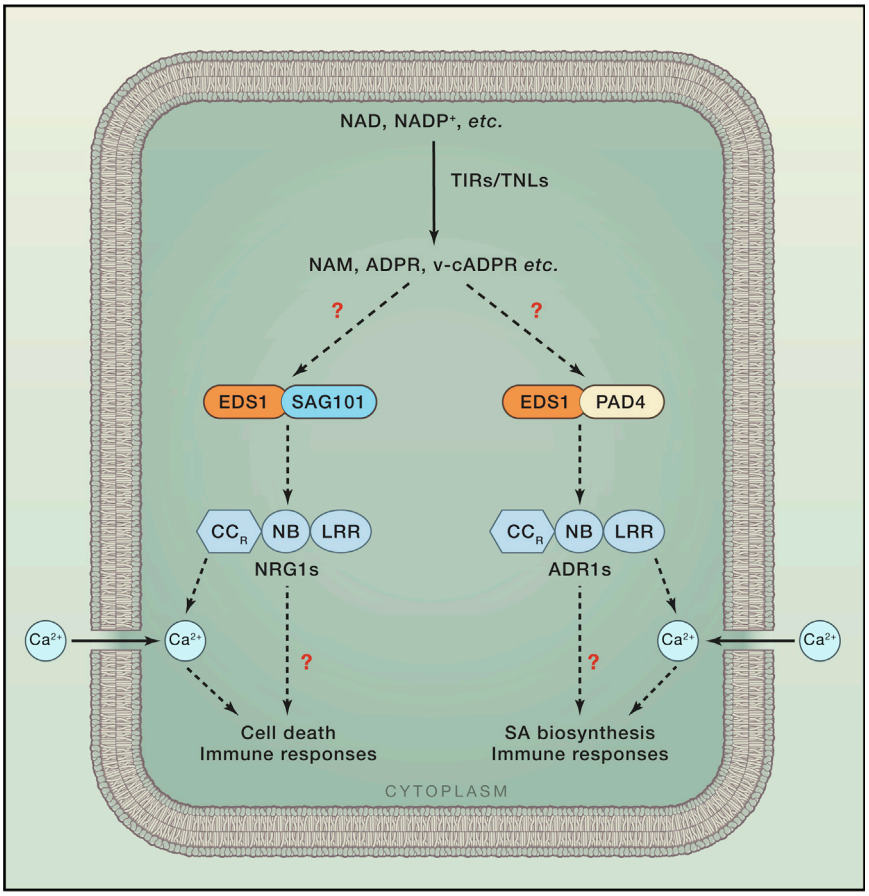



**Figure 3. A Working Model for TNL-Mediated Immune Signaling**

Upon activation, the TIR domains of TNLs cleave NAD, NAD phosphate (NADP+), and other related small molecules to generate NAM, ADPR, v-cADPR, etc. Some of these cleavage products serve as secondary signals to activate the EDS1/SAG101 and EDS1/PAD4 complexes and their downstream helper NLRs, NRG1s, and ADR1s. NRG1s promote cell death, whereas ADR1s induce SA biosynthesis, which are achieved by triggering $Ca^{2+}$ influx or through unknown mechanisms. Solid arrows are supported by literature, whereas dashed ones are hypothetical.

residues located in the outer surface of the ZAR1 barrel structure are not only conserved but also required for function of the newly analyzed MADA-containing CNLs. Importantly, this motif is functionally interchangeable, suggesting that the proposed pore-forming activity is involved in the function of many CNLs. However, it should be noted that some CNLs, such as RPS2, do not carry a canonical CC observed in ZAR1 and NRCs, and thus are unlikely to have a similar pore-forming function. The mode of action of these CNLs remains to be elucidated.

It is unclear whether the pore-forming activity of ZAR1 directly kills the cell. In animals, activated caspases cleave gasdermin-D, which matures into a pore-forming protein and oligomerizes into large pores of ~22 nm in diameter on the PM, leading to pyroptosis characterized by a swelling and lysis of the cell (Mulvihill et al., 2018; Shi et al., 2015). The ZAR1 pore is significantly smaller (6–10 Å in size) and might not kill the cell in a similar manner. Structural homology analyses suggest that CC and $CC_R$ domains of some plant NLRs adopt a fold similar to HeLo and HeLo-Like (HELL) domains of fungal proteins and four helical bundle animal mixed lineage kinase-like (MLKL) proteins (Daskalov et al., 2016; Jubic et al., 2019). Both MLKL and HeLo/HELL domain proteins can oligomerize to form a pore in the PM and cause necroptosis, a distinct type of programmed cell death (Daskalov et al., 2016). Although it remains unresolved as to how MLKL kills the cell, the MLKL pore has been reported to possess cation channel activity (Xia et al., 2016). Previous and recent reports showed that $LaCl_3$, a channel blocker, can inhibit cell death triggered by RPM1 and ZAR1 (Grant et al., 2000; Wang et al., 2019a). It is tempting to speculate that the ZAR1 resistosome pore might function as an ion channel (Figure 2), a possibility requiring future studies.

## Biochemical Functions of the TIR Domains

Increasing evidence supports the TIR domain of plant TNLs as a signaling module. The TIR-only RBA1 protein activates immune responses upon recognition of the bacterial effector protein HopBA1 (Nishimura et al., 2017). Transient overexpression of the TIR domains of some TNLs and a TIR-only protein from the monocot *Brachypodium distachyon* (BdTIR) is sufficient to trigger cell death in plants (Horsefield et al., 2019; Swiderski et al., 2009; Wan et al., 2019). Structural and functional analysis of several TNLs revealed that TIR domains form homodimers and heterodimers, and self-association of the TIR domains is required for defense signaling (Bernoux et al., 2011; Williams et al., 2014).

SARM1 is a non-canonical animal TIR protein acting as an NADase to cleave the oxidized form of NAD ($NAD^+$) (Essuman et al., 2017). TIRs of plant TNLs share similar structures as the SARM1 TIR (Horsefield et al., 2019). Interestingly, RBA1, BdTIR, as well as TIRs of several TNLs also exhibit $NAD^+$-cleaving activities upon dimerization, and the NADase activity is required for HR in plants (Horsefield et al., 2019; Wan et al., 2019). Taken together, pathogen effector-induced dimerization/oligomerization of TNLs facilitated by the NB domain could be a driving force to activate the NADase activity and immune signaling (Figure 3).

Although SARM1 and TIRs of plant TNLs both possess NADase activities, they induce cell death through distinct mechanisms (Wan et al., 2019). Enhanced disease susceptibility 1 (EDS1) is a general signaling component downstream of TNLs (Cui et al., 2015). EDS1 is required for cell death triggered by RBA1 and TIRs of TNLs but not for cell death induced by transient expression of an autoactive fragment of SARM1 in plants (Horsefield et al., 2019; Wan et al., 2019). SARM1 is believed

to trigger cell death by a depletion of $NAD^+$ (Essuman et al., 2017). Considering that the TIRs of most TNLs have very low $NAD^+$-cleaving activities (Horsefield et al., 2019; Wan et al., 2019), depletion of $NAD^+$ is unlikely to be a mechanism for cell death triggered by TNLs. Cleavage of $NAD^+$ by RBA1 and BdTIR generates several products including nicotinamide adenine mononucleotide (NAM), adenosine diphosphate ribose (ADPR), and a variant of cyclic ADP-ribose (v-cADPR) (Wan et al., 2019). These products or their derivatives could serve as signals to activate EDS1-dependent defenses (Figure 3).

## Signaling Downstream of TNLs

Signaling after EDS1 branches into two parallel pathways, which are dependent on SAG101 (senescence-associated gene 101) and PAD4 (phytoalexin deficient 4), respectively (Figure 3). Both SAG101 and PAD4 share a lipase-like domain with EDS1, and they form distinct complexes with EDS1 for the activation of downstream defenses (Cui et al., 2015). Although the mode of action of these complexes remains unknown, a cavity formed between the heterodimers could serve as a binding pocket for unknown proteins or ligands that activate immune signaling (Bhandari et al., 2019).

The *Nicotiana benthamiana* NRG1 is required for HR triggered by the N protein, a sensor TNL (Peart et al., 2005). Recent studies show that NRG1 function together with EDS1 and SAG101 in the activation of HR induced by TNLs such as the *N. benthamiana* Roq1 and *Arabidopsis* RPP1 (Gantner et al., 2019; Lapin et al., 2019; Qi et al., 2018). *Arabidopsis* has two redundant NRG1 orthologs, NRG1a and NRG1b, and both are required for HR and immunity triggered by various TNLs (Castel et al., 2019; Wu et al., 2019) (Figure 3). *Arabidopsis* RNLs ADR1, ADR1-L1, and ADR1-L2 act downstream of EDS1 and PAD4 to promote SA biosynthesis triggered by TNLs and some CNLs (Bonardi et al., 2011; Dong et al., 2016) (Figure 3). Both the EDS1/SAG101/NRG1 and EDS1/PAD4/ADR1 pathways contribute to resistance to pathogens. Simultaneous elimination of NRG1s and ADR1s leads to complete loss of TNL-mediated immunity (Wu et al., 2019). How defense signals are transduced through EDS1/PAD4 and EDS1/SAG101 to activate NRG1s and ADR1s remains to be determined (Figure 3).

A number of TNLs were reported to localize in the nucleus and associate with transcriptional regulators, suggesting that they are either activated by pathogen effectors in the nucleus or directly involved in transcriptional regulation of defenses (Li et al., 2015). Surprisingly, the *Arabidopsis* NRG1 localizes in the cytosol and partially co-localizes with endoplasmic reticulum (ER) and PM-markers, suggesting membrane-specific functions outside the nucleus (Wu et al., 2019). As discussed earlier, the $CC_R$ domain shares a similar structural fold with the four helical bundle domains of MLKL and the HeLo/HELL proteins (Jubic et al., 2019). It will be interesting to test whether activated ADR1s and/or NRG1s form pores on either the PM or ER for immune signaling (Figure 3). The spatial coordination between the TNLs and RNLs also warrant future examination.

## Different Classes of NLRs Are Organized in Networks

An emerging theme is that different NLRs form multi-layered networks in their functionality (Jubic et al., 2019; van Wersch and Li, 2019). Although some sensor NLRs referred to as "singletons" (such as RPM1, RPS2, and ZAR1) appear to act alone, others act in pairs to perceive effector proteins. For example, the *Arabidopsis* TNLs RRS1 and RPS4 form a complex, in which RRS1 acts as a sensor and RPS4 as a "partner" (Le Roux et al., 2015; Sarris et al., 2015). The TNL SNC1 is likely to physically pair with three redundant TNL partners, SIKIC1, SIKIC2, and SIKIC3, in immune activation (Dong et al., 2018). NRCs are slow evolving CNLs in Solanaceae plants. Each NRC functions downstream of multiple fast-evolving CNLs, although it is not known whether the upstream CNLs directly interact with NRCs (Wu et al., 2017). Furthermore, RNLs are required for function of TNLs and, in some cases, CNLs (Bonardi et al., 2011). Although NRCs, RNLs, and partner NLRs (such as RPS4 and SIKICs) have all been referred to as "helper NLRs" in the literature, their functions likely differ (see above). Here we use helper NLRs specifically for slow-evolving NLRs including NRCs and RNLs, in contrast with the fast-evolving sensor NLRs. Although NRCs and RNLs might directly execute immune signaling in a way similar to ZAR1 or through an unknown mechanism, partner NLRs RPS4 and SIKICs likely first act through the NADase activity. How upstream CNLs and TNLs are mechanistically linked to NRCs and RNLs and execute immune signaling require future investigation.

## SA Biosynthesis in Plant Immunity

Plants produce defense hormones ethylene, jasmonates, and SA to modulate defenses, among which SA plays essential roles in resistance against biotrophic and hemi-biotrophic pathogens. SA has profound importance in the amplification of PRR- and NLR-mediated immune signaling (Zhang and Li, 2019). Some major progress has recently been made in the understanding of SA biosynthesis (Huang et al., 2020). In *Arabidopsis*, pathogen-induced SA is primarily produced through the isochorismate pathway (Wildermuth et al., 2001). Isochoristmate synthase 1 (ICS1) converts chorismate into isochorismate in plastids. Recently it was shown that enhanced disease susceptibility 5 (EDS5) transports isochorismate from plastids to cytosol, where it is conjugated to glutamate by AVRPPHB susceptible 3 (PBS3) to form isochorismate-9-glutamate (Rekhter et al., 2019). Subsequently SA is produced via spontaneous decay of isochorismate-9-glutamate in a process which can be accelerated in the presence of enhanced pseudomonas susceptibilty 1 (EPS1) (Rekhter et al., 2019; Torrens-Spence et al., 2019).

Except *EPS1*, all the genes involved in pathogen-induced SA biosynthesis are strongly upregulated during plant defense. The expression of *ICS1* is under complex regulation that involves many positive and negative regulators (Huang et al., 2020). Two transcription factors, SAR deficient 1 (SARD1) and calmodulin binding protein 60-like g (CBP60 g), directly promote the expression of *ICS1*, *EDS5*, and *PBS3* (Sun et al., 2015; Zhang et al., 2010). As discussed in a recent review (Huang et al., 2020), $Ca^{2+}$ signaling also plays indispensable roles in activation of SA biosynthesis by modulating the activities of CBP60 g and CaM-binding transcription factor CAMTA3.

## Regulation of Defense Gene Expression by SA

Unlike most plant hormones, SA is perceived by two different classes of receptors, NPR1 and NPR3/NPR4, which play

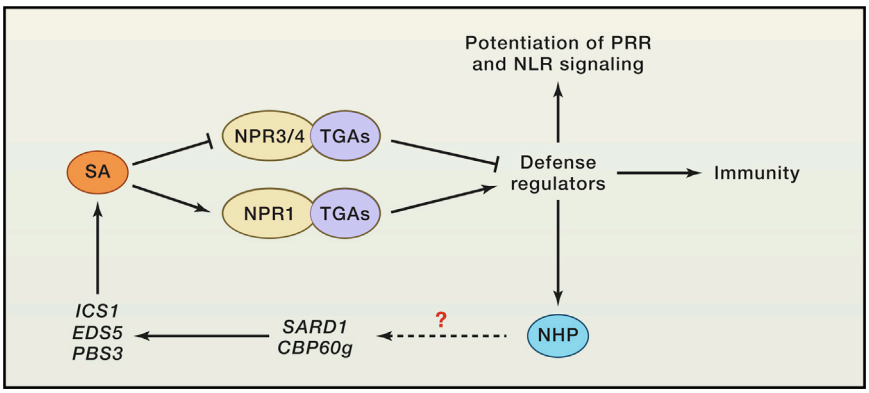



**Figure 4. A Working Model for SA and NHP-Mediated Immune Regulation**
Elevated SA levels induce defense gene expression by promoting the transcriptional activator activity of NPR1 and inhibiting the transcriptional repression activity of NPR3/NPR4. Both NPR3/4 and NPR1 partner with TGA transcription factors. NHP promotes SA biosynthesis by inducing the expression of *SARD1* and *CBP60 g* to prime plant defenses. Solid arrows are supported by literature, whereas dashed ones are hypothetical.

opposite roles in regulating defense gene expression (Zhang and Li, 2019). They are recruited to the promoters of SA-responsive genes by the redundant transcription factors TGA2, TGA5, and TGA6 (Figure 4). NPR1 functions as a transcriptional activator, whereas NPR3 and NPR4 serve as redundant transcriptional repressors. SA binding stimulates the transcriptional activation activity of NPR1 and inhibits the transcriptional repression activities of NPR3/NPR4, leading to induction of defense-related genes (Ding et al., 2018) (Figure 4).

Treatment with SA rapidly induces a large number of signaling components in the PRR pathway, among which are a number of PRRs and their co-receptors as well as many signaling components downstream of PRRs (Ding et al., 2018). SA treatment also induces the expression of some sensor NLRs and most known NLR pathway signaling components. This could potentiate PRR- and NLR-mediated immune responses, as supported by compromised PRR- and NLR-mediated immunity in mutants defective in SA perception or biosynthesis (Zhang and Li, 2019). In addition, SA is also involved in negative feedback regulation of HR triggered by NLRs, as indicated by enhanced cell death triggered by the CNL RPS2 in *Arabidopsis* mutants deficient in SA accumulation or perception (Radojičić et al., 2018). Thus, SA signaling is well-integrated into both PRR and NLR signaling pathways.

## Stimulation of SA Signaling by NHP

Recently, N-hydroxypipecolic acid (NHP) was identified as another important signaling molecule working along with SA in both local and systemic immunity (Chen et al., 2018; Hartmann et al., 2018). NHP is biosynthesized from lysine in three steps catalyzed by the aminotransferase ALD1, the reductase SARD4, and the monooxygenase FMO1, which are coordinately regulated by SARD1 and CBP60 g (Huang et al., 2020). Treatment of Pip/NHP transcriptionally primes SA biosynthesis, most likely by inducing *SARD1* and *CBP60 g*, although the underlying mechanism remains unknown (Hartmann et al., 2018; Chen et al., 2018) (Figure 4). Interestingly, two studies indicated that SA might also positively regulate NHP biosynthesis (Ding et al., 2018; Sun et al., 2020), suggesting mutual amplification between SA and NHP signaling.

## PRRs and NLRs at a Crossroad

Activation of NLRs lead to massive production of ROS, calcium influxes, activation of MAP kinases, defense gene expression, synthesis of defense hormones, and programmed cell death. Many of these responses are similar to those induced upon PRR activation, but notable differences exist. Activation of PRRs usually does not trigger cell death. Activation of ROS burst, calcium influx, and MAPK cascade by PRRsis transient in nature, whereas these responses after NLR activation are often delayed and prolonged (Peng et al., 2018; Su et al., 2018; Tsuda et al., 2013).

A recent phosphoproteomics study provided evidence for early convergence of the PRR and NLR signaling pathways (Kadota et al., 2019). By using transgenic plants conditionally expressing AvrRpt2, it was shown that RPS2 activation leads to increased phosphorylation of a number of immune signaling proteins such as RbohD and XLG2, many of which are also phosphorylated on the same sites during PRR signaling. How activation of transmembrane PRRs and intracellular NLRs results in similar phosphorylation changes on these sites is not understood. Some of them are known phosphosites of CPKs, which might be phosphorylated in response to changes in $Ca2^{+}$ levels after activation of PRRs or NLRs. Interestingly, two serine residues on RbohD known to be phosphorylated by BIK1 during FLS2 activation are also phosphorylated upon RPS2 activation.

Two most recent studies show that activation of NLRs by transgenic expression of cognate effectors in the absence of MAMP perception only modestly induces defenses (Ngou et al., 2020; Yuan et al., 2020). Surprisingly, simultaneous stimulation of PRRs and NLRs is required for full activation of defenses including ROS burst, callose deposition, prolonged MAPK activation, defense gene induction, HR, and complete disease resistance. In addition, pre-activation of NLRs also primes PRR-mediated defenses (Ngou et al., 2020), indicating intimate crosstalks between the PRR and NLR pathways. Much of the synergistic defense activation is associated with increased expression of genes encoding PRRs, MAPK cascades, RLCKs, G proteins, Rboh, CPKs, candidate calcium channel proteins, and proteins regulating SA and Pip biosynthesis, suggesting that the two pathways mutually potentiate each other through increased accumulation of signaling components and SA (Ngou et al., 2020) (Figures 2 and 4). However, the synergistic phosphorylation of RbohD at S343 and S347 can be uncoupled from increased expression of RbohD (Yuan et al., 2020), suggesting an early crosstalk between the PRR and NLR pathways. Furthermore, $Ca2^{+}$ could serve as a signal in such crosstalk. Calcium flux triggered by PRRs and NLRs could potentiate each other via RbohD, CPKs, CBP60 g, CAMTA3, or the release of

Peps and unknown phytocytokines for further activation of immunity (Figure 2). Future efforts are needed to decipher the complex interactions between the two pathways. Knowledge gained from PRR signaling studies could prove useful in unraveling NLR signaling.

## Concluding Remarks

Research from the last three decades has established a solid framework for the plant immune system. Studies on the PRR and NLR networks have highlighted the enormous complexity in danger perception, which likely requires integration of multiple immunogenic signals. RLCKs have emerged as key players in defense signaling, which connect PRRs to downstream calcium channels, NADPH oxidases, MAPK cascades, etc. Recent structural and functional analysis of NLRs has shed new light on the signaling mechanisms of the CC and TIR domains. Studies on the SA receptors NPR1 and NPR3/NPR4 have shed new light on how SA promotes defense gene expression and serves as a major signaling amplifier in PRR and NLR pathways. Meanwhile, recent completion of the SA and NHP biosynthesis pathways in *Arabidopsis* has allowed better understanding of their regulation and the crosstalk between these two critical defense signals. Despite the remarkable progress in plant immune signaling research, there are still many unanswered questions such as how different PRRs, accessory LRR-RKs, and scaffold proteins are coordinated during MAMP perception and immune activation; how RNLs and NRCs are activated by upstream TNLs to execute immune signaling; how NLR and PRR pathways interact; and how NHP is perceived by plants to stimulate *SARD1* and *CBP60 g* expression. Addressing these questions will fill the gaps in our understanding of the plant defense network. In addition, comparative studies of plant defense pathways across different plant species will lead to better understanding of the evolution of plant immunity.

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